

TECHNICAL REPORT

It's Not the Network.

Pinpointing Application vs. Network Issues

Live measurements with Prisma SD-WAN and Stigix — September 26, 2026

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Tests run with Stigix: <https://github.com/jsuzanne/stigix>

Executive summary

A branch user reports that one application is slow. The network team sees no alert, SD-WAN tunnels are green, and a ping to the data center returns 6 ms. Is it the network? Is it the application? Without the right tooling, that question takes hours to answer.

The answer in 30 seconds. Stigix and Prisma SD-WAN both measured the Server Response Time (SRT) of the slow application while confirming that the network transport was healthy. The result was unambiguous: the SD-WAN was delivering packets in under 10 ms. The server was taking 800 ms to respond. The network was innocent.

5.8 ms Network RTT (convergence probe)	99% of app delay caused by the server	0% packet loss during the test	< 30 s time to reach a verdict
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This report documents the test, the measurements from both Stigix and Prisma SD-WAN SCM, and the diagnostic method. A counter-example — a real network degradation injected with the same tooling — shows what a network problem looks like by contrast.

Important scope note

SRT (Server Response Time) measurement in Stigix is available for **HTTP/REST applications** (such as Onprem8096, configured in HTTP/1.1 REST mode). For plain TCP applications without an HTTP layer, Stigix measures the application RTT but does not decompose the server delay directly. Prisma SD-WAN SCM provides SRT visibility for all recognized applications through its own telemetry, as shown in the SCM dashboards in this report.

Test setup

Branch BR8 runs two persistent workloads toward data center DC1, both through the same Prisma SD-WAN overlay (tunnel BR8-INET2 → DC1-INET):

Application	Port	Protocol	Server behavior	Expected SRT
Onprem8083	8083	HTTP/1.1 REST	Echo — replies immediately	≈ network RTT only
Onprem8096	8096	HTTP/1.1 REST	Fixed Delay 800 ms before every reply	800 ms + network RTT

Onprem8083 is the healthy reference. Onprem8096 simulates a slow backend — a delayed database query, an overloaded API, or a misconfigured timeout. The network is not modified.

Server Role / Inbound (Incoming Request Handling)

These settings define how **this node replies** when receiving requests on its local listener port (:8096). *If this node only acts as a Client sending traffic to a remote Data Center, this behavior remains dormant locally (the behavior configured on the remote target server will execute).*

Server Response Simulation Mode

Echo

Replies back with exact payload

Acknowledge

Compact ACK without payload

Fixed Delay

Injects fixed delay before reply

Random Delay

Jittered delay between min/max

Looping Delay

Alternates normal/slow phases

Drop Response

Simulates response loss

Close Connection

Closes socket after N requests

Error Response

Simulates server application errors

EICAR Security Test

Emulates malware payload for AV/NGFW

Simulation Parameters (fixed_delay)

Fixed Delay (ms)

800

Fixed Delay Simulation

Injects a constant synthetic delay on every server response frame. *Simulates predictable high-latency WAN, transcontinental links, or slow backend database processing.*

Back

Next Step

Figure 1. Stigix app configuration: server behavior set to Fixed Delay 800 ms. This injects a constant server processing delay on every HTTP response.

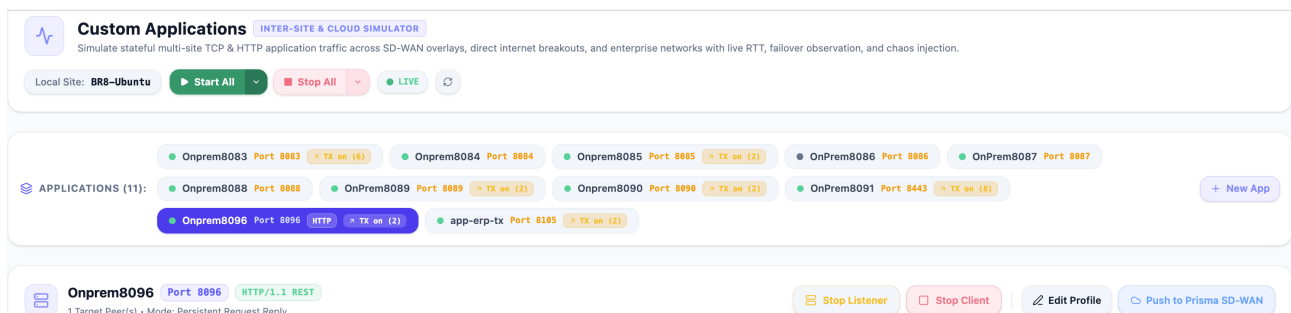


Figure 2. Stigix custom apps dashboard on BR8. Onprem8096 (port 8096) shows 855 ms average App RTT, while Onprem8083 and other apps are healthy. The contrast is immediately visible.

Results

State 1 — Healthy reference: Onprem8083

Metric	Value	Source
Network RTT (convergence probe)	5.8 ms — PERFECT	CONV-0270
App RTT p50 → DC1	10 ms	Stigix get_tcp_app_sessions
serverDelayMs (HTTP)	0 ms — server replies instantly	Stigix SRT breakdown
Voice MOS → DC1	4.4 / 5.0	Voice simulation
Packet loss	0 %	Convergence probe

State 2 — Slow application: Onprem8096 (800 ms fixed delay)

Metric	Value	What it means
Network RTT (convergence probe)	5.8 ms — PERFECT	Network unchanged
App RTT p50 → DC1	819 ms	80× slower than the healthy reference
App RTT p95 → DC1	1 271 ms	Tail latency at peak load
serverDelayMs (HTTP)	800 ms → 99% of total RTT	Server is the bottleneck
networkRttMs	68 ms → 1% of total RTT	Network is not the issue
Voice MOS → DC1	4.4 / 5.0	Completely unaffected
Timeouts / errors	0	Sessions remain stable
Packet loss	0 %	Network is healthy

**DC1-Ubuntu**

OUTGOING SESSION



local → 192.168.203.100:8096 • Onprem8096

● **CONNECTED** | ⌚ Uptime: **4h 21m**Jitter: ± **75.92 ms**⚡ **Latency Breakdown (RTT)**

🔄 Flow & Bitrates

📄 Socket & Protocol

⚡ **LATENCY ORIGIN BREAKDOWN****Total App RTT: 810 ms**● **SD-WAN Transport: 10 ms (1%)**● **Server Processing Delay: 800 ms (99%)**

⚠ **Simulated Server Delay Active:** The remote server behavior adds a fixed or random artificial processing delay of **800ms**. The true network path transit is **10ms**.

MINIMUM

806 ms

MEDIAN (P50)

817 ms

TAIL (P95)

1094 ms

PEAK (MAX)

1570 ms📊 **Real-Time RTT Waveform (Last 20 ticks)**

Avg: 860.15 ms

📄 **Understanding Latency Metrics & Percentiles**

Median (P50): 50% of requests are faster than this value. Represents typical baseline performance.

Minimum: Best observed physical transport floor across underlay/overlay tunnels.

Tail Latency (P95): 95% of requests complete within this delay. Highlights SD-WAN tunnel jitter and micro-burst queuing.

Peak (Max): Worst-case outlier transaction recorded in the sliding window.

Real-time telemetry stream active (1.5s refresh)

⚡ **Send SYN/Handshake Probe****Close Drawer**

Figure 3. Stigix Latency Origin Breakdown (HTTP app): SD-WAN Transport 10 ms (1%) vs Server Processing Delay 800 ms (99%). The network is innocent.

Prisma SD-WAN SCM confirms the same picture

Prisma SD-WAN's own analytics platform, SCM, independently surfaces the same diagnosis. The application STX_Onprem8096 is pushed to Prisma SD-WAN via Stigix, making its SRT metrics visible in the standard SCM dashboards alongside any other enterprise application.

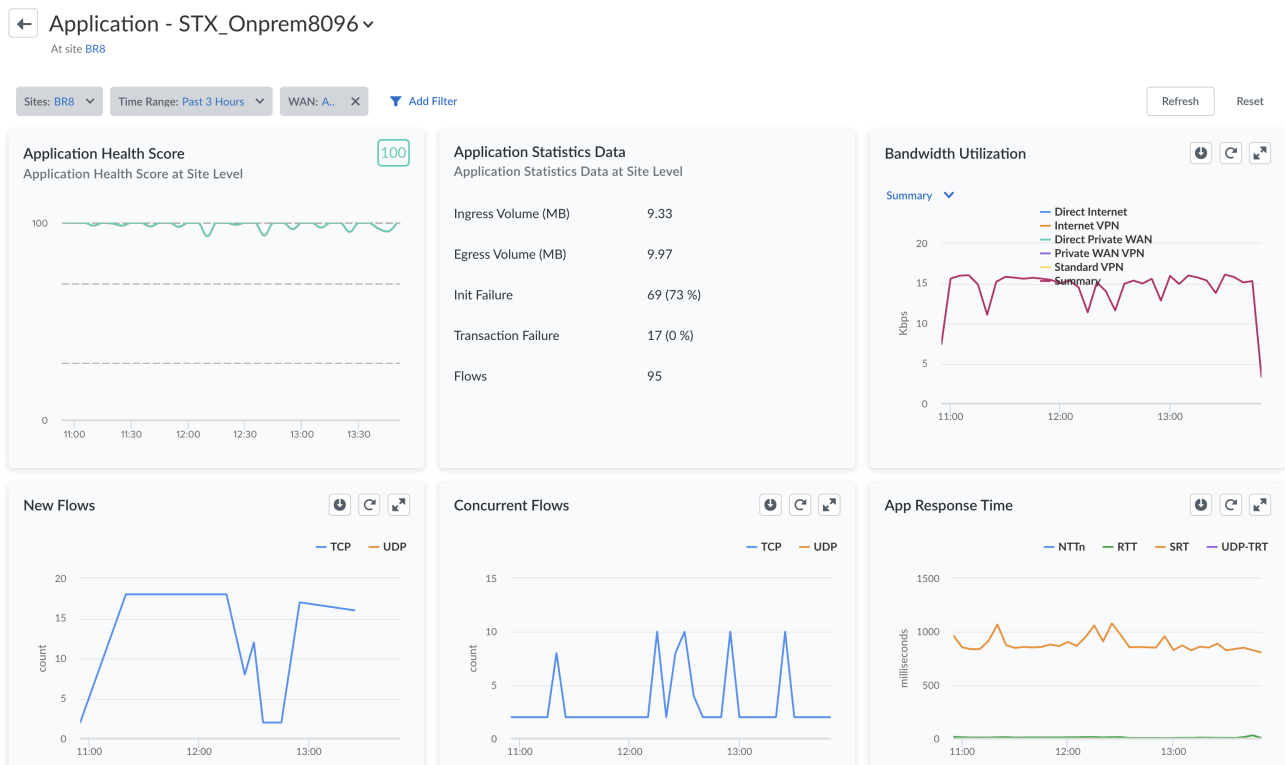


Figure 4. Prisma SD-WAN SCM — Application dashboard for STX_Onprem8096 at site BR8. Application Health Score: 100/100. App Response Time (SRT) visibly elevated in orange at ~850–1 000 ms. RTT (green) stays flat near 0. The network path is healthy; the server response time is the anomaly.

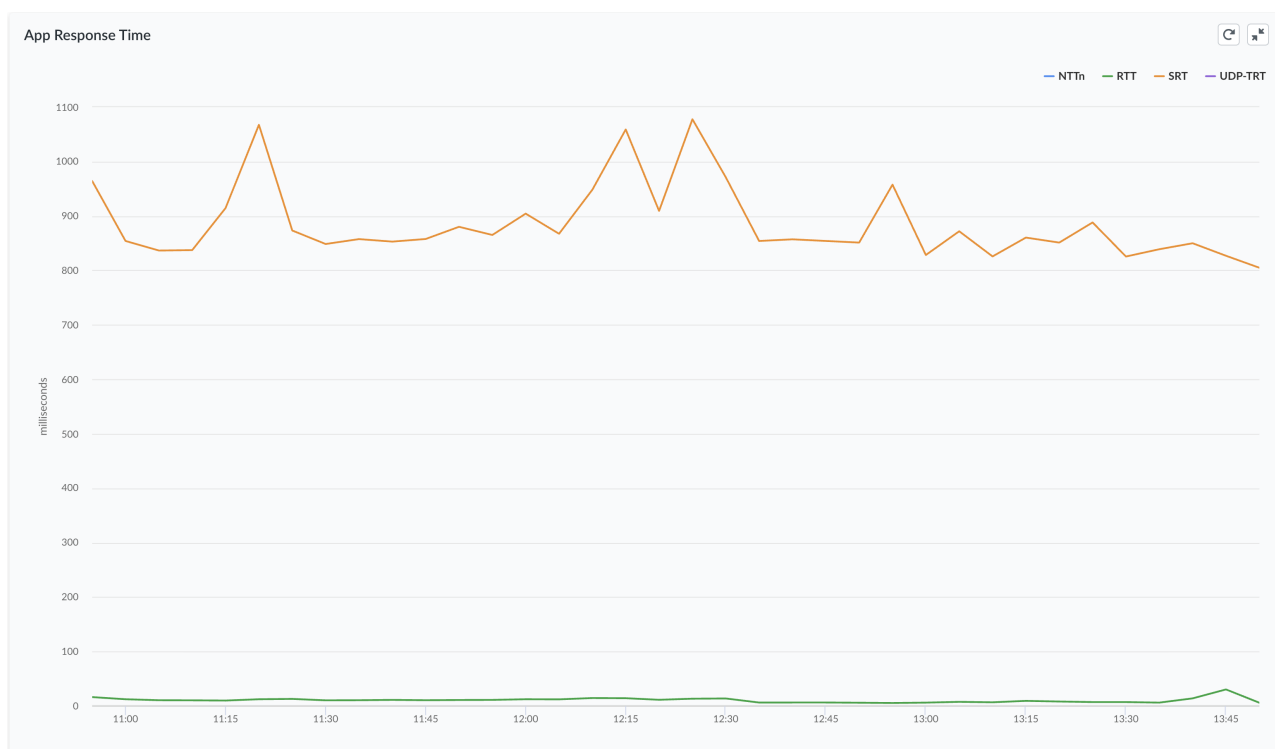


Figure 5. SCM App Response Time detail: SRT (orange) consistently at 800–1 100 ms over 3 hours. RTT (green) stays below 20 ms throughout. This is the Prisma SD-WAN native view of the same problem Stigix measured.

Flow Browser: path confirmed healthy

The Prisma SD-WAN Flow Browser confirms that the slow application used the correct path throughout the test, with no path changes and no SD-WAN routing anomaly:

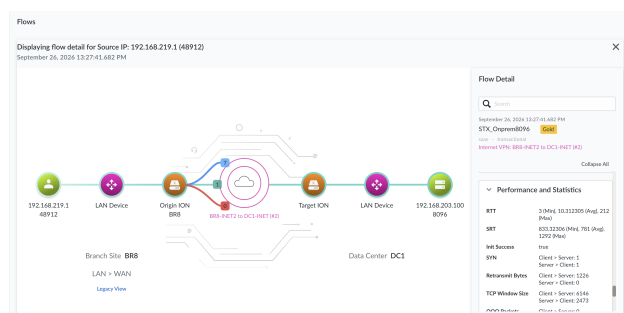


Figure 6. Flow Browser: STX_Onprem8096 (port 8096) traveling BR8-INET2 → DC1-INET (#2). Path stable, Init Success: true.

Performance and Statistics

RTT	3 (Min), 10.312305 (Avg), 212 (Max)
SRT	833.32306 (Min), 781 (Avg), 1292 (Max)
Init Success	true
SYN	Client > Server: 1 Server > Client: 1
Retransmit Bytes	Client > Server: 1226 Server > Client: 0
TCP Window Size	Client > Server: 6146 Server > Client: 2473

Figure 7. Flow statistics: RTT avg 10 ms, SRT avg 781 ms. The gap between RTT and SRT is the server processing delay — visible directly in SCM.

Two independent tools, same conclusion. Stigix measured serverDelayMs = 800 ms and networkRttMs = 68 ms from the LAN side. Prisma SD-WAN SCM measured SRT = 781 ms (avg) and RTT = 10 ms from the overlay. Both point to the same root cause: the server, not the network.

Counter-example: what a real network problem looks like

To validate the diagnostic method, +200 ms of latency was injected on eth10 (DC1-INET-221), the underlay circuit carrying the SD-WAN tunnel. The contrast with the slow-application case is immediate:

Signal	Slow app (Onprem8096)	Network degradation
Convergence probe RTT	5.8 ms — PERFECT	93 ms — degraded
Apps affected	Onprem8096 only	All apps toward DC1 simultaneously
serverDelayMs	800 ms	800 ms — unchanged
networkRttMs	68 ms	238 ms — +250%
Voice MOS	4.4 — unaffected	Degraded
Packet loss	0 %	0.4%, blackout 183 ms
Diagnosis	Server / application	Network / transport

The two signatures are unmistakable: a slow application affects only that application, leaves the convergence probe unchanged, and shows a large serverDelayMs. A network degradation affects all applications simultaneously, climbs the probe, and leaves serverDelayMs constant.

The diagnostic in two checks

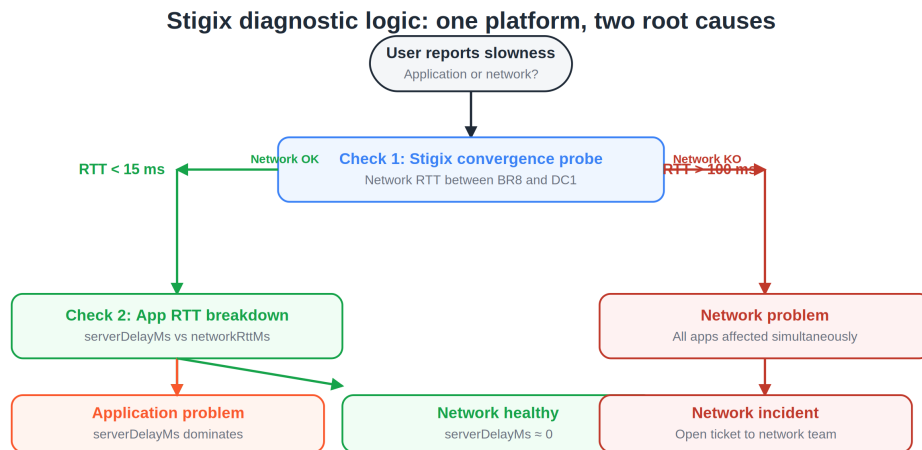


Figure 8. Diagnostic flow: two Stigix checks lead to an unambiguous verdict in under two minutes.

- Check 1 — Convergence probe:** is the network RTT between the branch and the data center normal? If it is high or shows loss, stop here: it is a network incident.
- Check 2 — App RTT breakdown (HTTP apps):** is *serverDelayMs* the dominant component? If yes, the server is the bottleneck. Verify with Prisma SCM's SRT metric for additional confirmation, including for non-HTTP applications.

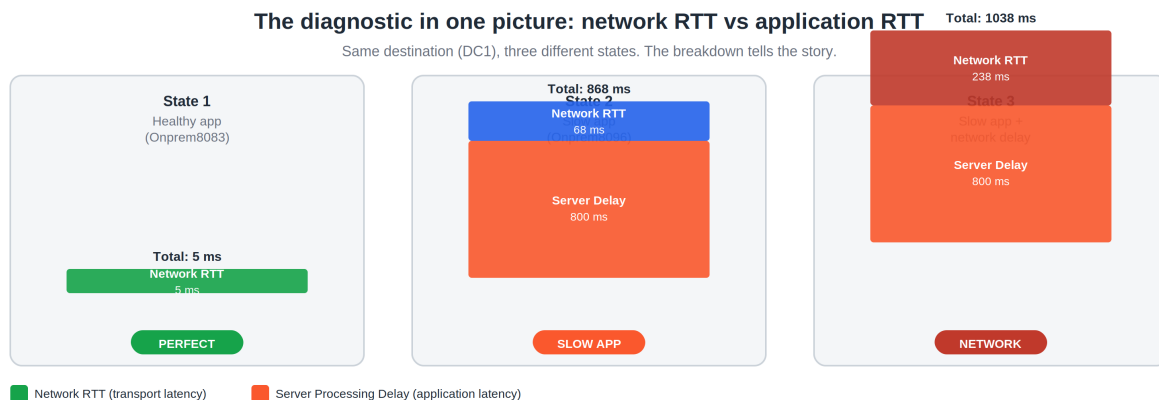


Figure 9. Three states, same destination. The breakdown of each bar tells the story.

Conclusions and recommendations

- 1 Don't stop at the ping.** A 6 ms ping says nothing about whether the server behind it responds in 10 ms or 800 ms. Use the App RTT breakdown for HTTP apps, and Prisma SCM's SRT metric for all applications.
- 2 Use both Stigix and Prisma SCM.** Stigix gives detailed session-level data from the LAN side. Prisma SCM gives SRT visibility for all recognized applications at the overlay level. Together, they cover every layer.
- 3 Run the counter-example.** Showing what a network problem looks like — all apps affected simultaneously, probe climbing, serverDelay stable — makes the diagnostic credible to both the network and application teams.
- 4 Set a baseline.** Record healthy SRT values for each critical HTTP application. Alert when SRT deviates by more than 3×.
- 5 Voice MOS is a free network health check.** If MOS stays at 4.4 while an application is slow, the network is almost certainly not the cause.

Measurement notes

SRT breakdown (serverDelayMs / networkRttMs) is available in Stigix for HTTP/REST applications. For plain TCP applications, Stigix measures the total App RTT; use Prisma SCM for SRT decomposition.

The 68 ms average networkRttMs during the test differs from the 5.8 ms probe RTT because it is averaged over the full session lifetime including connection setup.

The +200 ms counter-example was applied on the underlay circuit (eth10), not the overlay. The convergence probe saw 93 ms — not 200 ms — because the SD-WAN had two underlay paths.